PPPx User Manual

Version 1.2.4

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Contents

1 Introduction

1.1 Overview

PPPx is a software package developed for multi-GNSS data processing. This manual covers its main component, `pppx`, which is dedicated to multi-GNSS positioning. Its primary features are listed below:

1. Supports four solution modes:
   (a) SPP: Single Point Positioning
   (b) PPP: Precise Point Positioning
   (c) RTK: Real-Time Kinematic (short baseline) positioning
   (d) TDP: Time-Differenced Positioning
2. Supports multi-GNSS: GPS, GLONASS, Galileo, BeiDou-2/3, and QZSS
3. Supports multiple solvers: Extended Kalman Filter (EKF), Factor Graph Optimization (FGO), and Least-Squares Estimator (LSQ)
4. Supports PPP-AR with Observable-specific Signal Bias (OSB) products
5. Flexible frequency selection (L1/L2/E1/E5a/...)
6. High precision and efficiency: capable of processing 2880 epochs within 1–2 s
7. Unified input/output for different solution modes
8. Supports Linux, Windows, and macOS

1.2 License

This software is distributed under the GPL-3.0 license.

2 Getting started

This section introduces the system requirements and installation steps for `pppx` on the supported platforms (Linux, macOS, and Windows).

2.1 System requirements

Generally, there is no specific hardware requirement for `pppx`. To process a 24-hour GNSS observation file with a sampling rate of 30 s, `pppx` consumes around 150 MB of resident memory and one CPU core. The resident memory consumed may, however, vary with the number of GNSS systems used and the data sampling rate.

Note that `pppx` only supports Macs with Apple silicon (i.e., ARM architecture) and a macOS version newer than Monterey.

2.2 Installation

2.2.1 Linux

This application is distributed as an AppImage, making it portable across most modern Linux distributions. Specifically, `pppx` is built on Ubuntu 20.04 and requires the host system to have GNU C Library (`glibc`) version 2.31 or newer.

It is compatible with the following distributions (and any of their derivatives):

Distribution Family	Supported Versions
Ubuntu	20.04 LTS, 22.04 LTS, 24.04 LTS (and newer)
Linux Mint	20, 21, 22 (and newer)
Debian	11 “Bullseye”, 12 “Bookworm” (and newer)
Fedora	32 (and newer)
RHEL / Rocky / Alma	9.x (and newer) *Note: RHEL 8 is not supported*
openSUSE	Leap 15.3+, Tumbleweed
Arch Linux / Manjaro	Rolling releases (Always updated)

Not sure if your system is supported?

You can easily check your system’s `glibc` version by running the `ldd -version` command in your terminal.

2.2.1.1 Option 1: Automated Installation (Recommended)

Run this single command to automatically download and install the latest release to `~/.local/bin`:

```
curl -sL https://raw.githubusercontent.com/YuanxinPan/PPPx_bin/main/install.sh | bash
```

2.2.1.2 Option 2: Manual Installation

If you prefer not to use the automated script, you can install pppx manually:

1. Download the `pppx_v1.x.x_linux_x86_64.zip` asset from the latest Release.
2. Extract the archive to obtain the pppx executable.
3. Open your terminal and navigate to your download folder.
4. Make the file executable and move it to your user binary folder:

```
chmod +x pppx
mkdir -p ~/.local/bin
mv pppx ~/.local/bin/pppx

# Restart your terminal and run `pppx` to test if it works
# If not, add `~/.local/bin` to PATH via the command below:
echo "export PATH=\${HOME}/.local/bin:\$PATH" >> ~/.bashrc
```

2.2.2 Windows

The easiest way to run pppx on Windows is to use the Windows Subsystem for Linux (WSL); simply follow the instructions in Section 2.2.1. However, pppx can also run natively on Windows using PowerShell or Command Prompt (`cmd.exe`).

2.2.2.1 Option 1: Automated Installation (Recommended)

Open PowerShell and run this command. It will automatically download the executable, fetch the required DLLs, and add the software (`C:\Users\<YOUR_USER_NAME>\AppData\Local\Programs\PPPx`) to your system `PATH`:

```
irm https://raw.githubusercontent.com/YuanxinPan/PPPx_bin/main/install.ps1 | iex
```

Note: You must restart PowerShell or cmd.exe after the installation.

2.2.2.2 Option 2: Manual Installation

If you prefer not to use the automated script, you can install pppx manually:

1. Download `pppx_v1.x.x_windows_x86_64.zip` from the latest Release.
2. Download the required Windows DLLs via this link.

3. Create the PPPx folder: `C:\Users\<YOUR_USER_NAME>\AppData\Local\Programs\PPPx`
4. Extract both archives and place the `.dll` files and `pppx.exe` in the PPPx folder.
5. Add the absolute path of this folder to your Windows `PATH` environment variable.
6. Open PowerShell or Command Prompt and type `pppx.exe` to test if it is correctly installed.

2.2.3 macOS

pppx only supports Macs with Apple silicon and a macOS version newer than Monterey.

2.2.3.1 Prerequisite

If you do not have Homebrew on your Mac, please install it first:

```
$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

pppx is linked against the latest `ceres-solver` library, so make sure you have the latest version installed as well in order to run pppx properly on your Mac. Run the following command in the Terminal application:

```
brew install ceres-solver
```

2.2.3.2 Option 1: Automated Installation (Recommended)

Run this command in your Terminal to automatically download the latest release, bypass Apple's quarantine security check, and install it to `~/.local/bin/`:

```
curl -sL https://raw.githubusercontent.com/YuanxinPan/PPPx_bin/main/install.sh | bash
```

2.2.3.3 Option 2: Manual Installation

If you prefer not to use the automated script, you can install pppx manually:

1. Download the `pppx_v1.x.x_macos_arm64.zip` asset from the latest Release.
2. Extract the archive to obtain the pppx executable.
3. Open the Terminal application and navigate to your download folder.
4. Move the pppx executable to `~/.local/bin/` using Terminal:

```
chmod +x pppx
mkdir -p ~/.local/bin
mv pppx ~/.local/bin/

echo "export PATH=\${HOME}/.local/bin:\$PATH" >> ~/.zshrc
```

5. Due to Apple's Gatekeeper security settings, you must authorize the software before it will run. You can do this by running `xattr -d com.apple.quarantine ~/.local/bin/pppx` or by following this Apple Support Guide.
6. Restart your Terminal and run `pppx` to test if it is correctly installed.

3 Usage

This section introduces how to use the pppx software. The general steps to run pppx are:

1. Prepare input files, including a GNSS observation file in the Receiver Independent Exchange (RINEX) format and necessary products (either broadcast ephemeris or precise products from IGS)
2. Modify the configuration file *pppx.ini* if necessary
3. Execute pppx from the command line:
   `pppx -c pppx.ini path-to-rnxobs [rnxobs-of-base]`
4. Check output files

Here are some example commands to run pppx:

```
# SPP/PPP/TDP
# (Note: TDP is suitable for high-frequency data, e.g., 1 Hz)
$ pppx -c pppx.ini rinex/ZIM200CHE_R_20221000000_01D_30S_MO.rnx

# RTK
# (ZIMM is the base station in this example)
$ pppx -c pppx.ini rinex/ZIM200CHE_R_20221000000_01D_30S_MO.rnx
↪ rinex/ZIMM00CHE_R_20221000000_01D_30S_MO.rnx
```

To make GNSS data processing easier, a wrapper script (scripts/pppx.sh) is provided that automatically downloads the necessary products (e.g., SP3, CLK, OBX, BIA, ERP, BRDC, and GIM) from the CODE FTP server and then invokes pppx. The recommended steps to use this wrapper script are:

```
$ pppx -x ppp > pppx.ini  # generate a default config file

# Then, edit pppx.ini if necessary but keep [product] (except "src")
↪ and [table] blank
$ pppx.sh -c pppx.ini rinex/ZIM200CHE_R_20221000000_01D_30S_MO.rnx
```

NOTE: Before first use, "scripts/pppx.sh" must be edited manually, setting the variable "TABLE_DIR" (line #42) to the actual "table" folder of the pppx software package.

3.1 Input

The complete list of input files is provided below:

- GNSS observations in the RINEX format, specified by command line arguments
- Configuration file `pppx.ini`, specified by command line argument
- Satellite products (either brdc or precise), specified by `pppx.ini`

- Table files (already provided in the table/ directory), specified by `pppx.ini`

The configuration file and satellite products will be explained in detail in subsequent sections.

3.1.1 Configuration

The configuration file `pppx.ini` is the primary interface to pppx. Written in the human-readable `ini` format, it lets users specify the GNSS positioning strategy through the sections listed in Table 3.1.

Table 3.1: Available sections of the configuration file `pppx.ini`.

Section	Description
session	The desired sampling rate and start and end time for data processing
constellation	The GNSS systems to be used and specific satellites to be excluded
observation	Selection of signal frequency and observable priority
model	The tropospheric and ionospheric models
estimation	Various estimation strategies, including solution mode, positioning mode, solver type, etc.
product	The choice of satellite product source and the path to each product
table	The paths to table files
output	The output path and log level

Some example configuration files are provided in the folder "example/pppx/". Most settings in `pppx.ini` are self-explanatory. Usually, users just need to modify the following settings and leave the rest at their default values:

```
[constellation]: system
[estimation]: sol_mode, pos_mode, solver
[product]: src, nav, sp3, clk
```

If no template configuration file is available, you can generate a default configuration for a specific solution mode (spp, ppp, rtk, or tdp) with the command below. Redirect its output to a file to obtain a ready-to-edit configuration for processing data with pppx.

```
# Output the default configurations for SPP and redirect them to a file
$ pppx -x spp > spp.ini
```

The following paragraphs explain how to set the key–value pairs in each section of the configuration file.

3.1.1.1 session

Here, "interval" specifies the processing interval; it must be no smaller than the sampling rate of the RINEX file and an integer multiple of it. "date" denotes the year and day of year (DOY) for data processing and is used to replace the placeholders in the "[product]" section. If left blank, it is read from the first epoch of the RINEX file. "start" and "end" denote the start and end epochs for data processing within the day (closed interval).

Figure 3.1 shows example settings. In most cases, all fields in this section can be left blank, and pppx will then process the entire RINEX file epoch by epoch.

```
[session]
interval = 300                    ; interval (sec)     [ RINEX ]
date     = 2020 100               ; year DOY           [ RINEX ]
start    = 02 15 30               ; hour min sec       [ none ]
end      = 23 45 00               ; hour min sec       [ none ]
```

Figure 3.1: The "session" section of configuration.

3.1.1.2 constellation

This section defines the GNSS systems to be used for data processing and the problematic satellites to be excluded. An example is shown in Figure 3.2. Note that the order of the letters (e.g., GRECJ) does not matter, and there must be no spaces between them. The GPS system is not mandatory; Galileo-only solutions, for example, are also feasible. The excluded PRNs should be separated by whitespace.

```
[constellation]
system  = GE                      ; opt: GRECJ         [ GRECJ ]
exclude = E14 E18 G04             ; excluded PRNs      [ none ]
```

Figure 3.2: The "constellation" section of configuration.

3.1.1.3 observation

This section is documented thoroughly by the comments in the configuration file. Note that the observation noise represents the noise level of the uncombined observations; for ionosphere-free combinations, the observation noise is computed automatically from the uncombined values. A screenshot of the example settings is provided in Figure 3.3.

```
[observation]
noise = 0.3 0.002                 ; observation noise of code/phase (m) [ 0.3 0.002 ]
; G/R/E/C/J = f1 f2 obs_priority: high -> low
G = 1 2 WPCLSXYMN                 ; default: [ 1 2 WPCLSXYMN ]
R = 1 2 PCIQX                     ; default: [ 1 2 PCIQX ]
E = 1 5 XCIQB                     ; default: [ 1 5 XCIQB ]
C = 2 6 IQX                       ; default: [ 2 6 IQX ]
J = 1 2 SLXCZ                     ; default: [ 1 2 SLXCZ ]
```

Figure 3.3: The "observation" section of configuration.

3.1.1.4 model

This section specifies the tropospheric and ionospheric models. An example is provided in Figure 3.4. Users should choose an option (case sensitive) from the values listed in the comments. Note that this section has no effect for RTK, since tropospheric and ionospheric delays are ignored for short baselines; "iono" can only be IF for PPP.

Currently, the tide model cannot be configured through `pppx.ini`. Solid Earth tide, ocean tidal loading, and pole tide are applied for PPP whenever possible, but not for spp/rtk/tdp.

```
[model]
trop = GMF                          ; opt: GMF/VMF1/GPT2w/none      [ GMF ]
iono = IF                           ; opt: IF/brdc/IONEX/none       [ IF ]
```

Figure 3.4: The "model" section of configuration.

3.1.1.5 estimation

This is the most important section of the configuration file. Here, you can choose the solution mode (spp/ppp/rtk/tdp), positioning mode (kinematic/static/fixed), and solver (ekf/fgo/lsq) used for adjustment. An example is provided in Figure 3.5. Most settings are self-explanatory.

The fields "sol_mode", "pos_mode", and "solver" are well explained in the configuration file. Note, however, that spp and tdp support only the kinematic positioning mode, and not every solver is supported for each solution mode (lsq and fgo for spp and tdp; ekf and fgo for ppp and rtk). The software will output an error message if a given configuration is not supported.

"fgo_var" controls the output of variance information when fgo is specified as the "solver". By default, variances are not computed for fgo, because doing so can be time-consuming.

"site_pos" specifies the approximate station coordinates in XYZ format (unit: meters). If not specified, the approximate coordinates are computed automatically using SPP. The same rule applies to the field "base_pos" for rtk.

"ppp_ar" is a switch for PPP-AR. Ambiguity resolution is enabled by setting it to "yes", in which case a valid phase-bias product must be provided through the field "[product] bia". The LAMBDA and rounding methods are used to resolve integer ambiguities when "solver" is set to "ekf" and "fgo", respectively. Note that GLONASS is disabled for AR and that the GPS L5 signal is currently not supported for PPP-AR.

"sf_ppp" is designed to run single-frequency PPP with GIM products; that is, ionospheric delays are not estimated but fixed to GIM corrections. This makes it possible to evaluate the accuracy of GIM products. Note that dual-frequency observations are still required for cycle slip detection, and "ppp_ar" should be set to "no" when sf_ppp is enabled.

"slip_det" specifies the methods used for cycle slip detection in ppp. The option "TDCP" stands for time-differenced carrier phase. You can set "slip_det" to "ALL" for most stations, but "MW" should not be used for stations with noisy pseudorange, such as smartphones. "LLI" can be unreliable, because some stations set many incorrect LLI flags, yet it is recommended for processing smartphone GNSS data. Note that rtk and tdp always use LLI and GF for cycle slip detection, which is currently not configurable.

"pos_pri", "clk_pri", "isb_pri", and "ztd_pri" specify the a priori uncertainties and process noises of the station position, receiver clock, receiver inter-system bias (ISB), and tropospheric delay for EKF, respectively. Note that EKF estimates receiver clocks for one reference GNSS system and estimates ISBs for other systems. When FGO is chosen as the solver, only "ztd_pri" is effective and the other "pri" settings are not used: an individual receiver clock error (instead of an ISB) is estimated as white noise for each constellation, and the station position is estimated as white noise in the kinematic mode.

"rtk_ar" is a switch for RTK-AR, with "yes" as its default value. The LAMBDA and rounding methods are used to resolve integer double-differenced (DD) ambiguities when "solver" is set to "ekf" and "fgo", respectively. "glonass_ar" can be set to "yes" for rtk if the rover and base stations share the same receiver type; otherwise, keep it as "no".

```
[estimation]
sol_mode    = ppp                    ; opt: spp/ppp/rtk/tdp         [ spp ]
pos_mode    = kinematic              ; opt: kinematic/static/fixed  [ kinematic ]
solver      = ekf                    ; opt: ekf/fgo/lsq             [ ekf ]
weight_opt  = elev                   ; opt: elev/snr                [ elev ]
elev_mask   = 7                      ; elevation mask (°)           [ 10 ]
snr_mask    = 25                     ; SNR mask (dB-Hz)             [ 25 ]
site_pos    =                        ; initial position in xyz (m)  [ spp ]
; ppp only
ppp_ar      = no                     ; PPP ambiguity resolution     [ no ]
sf_ppp      = no                     ; single-frequency PPP         [ no ]
slip_det    = ALL                    ; opt: off GF MW LLI TDCP ALL  [ ALL ]
pos_pri     = 100 1                  ; uncertainty  process_noise   [ 1E+02 1 ]
clk_pri     = 100 100                ;      m          m/sqrt(s)    [ 1E+02 1E+02 ]
isb_pri     = 50  3.2E-04            ;                              [ 5E+01 3.2E-04 ]
ztd_pri     = 0.5 3E-05              ;                              [ 0.5 3E-05 ]
; rtk only
rtk_ar      = yes                    ; RTK ambiguity resolution     [ yes ]
glonass_ar  = no                     ; AR for GLONASS               [ no ]
base_pos    =                        ; base position in xyz (m)     [ spp ]
```

Figure 3.5: The "estimation" section of configuration.

3.1.1.6 product

This section specifies the paths to the various products. If "src = brdc", then "nav" must be set to a valid path to a broadcast ephemeris. If "src = precise", then "sp3" must be set to a valid path to an SP3 product, and all other products are optional. Relative paths are resolved relative to the directory where pppx is executed, not where pppx.ini is located. In addition, placeholders (see Figure 3.6) can be used in product names, making it easier to process GNSS data over multiple days. More details on the products are given in Section 3.1.2.

```
; available placeholder:
;   -YEAR-   : year
;   -YR-     : 2-digit year
;   -DOY-    : day of year
;   -WEEK-   : GPS week
;   -DOW-    : day of GPS week
[product]
src = precise                                        ; opt: brdc/precise
nav = rinex/BRDC00IGS_R_-YEAR--DOY-0000_01D_MN.rnx   ; broadcast ephemeris        (brdc)
sp3 = products/cod-WEEK--DOW-.sp3                    ; precise oribt              (precise)
clk = products/cod-WEEK--DOW-.clk                    ; precise clock              (precise)
erp = products/cod-WEEK--DOW-.erp                    ; earth rotation parameters  (precise)
bia = products/cod-WEEK--DOW-.bia                    ; satellite bias             (precise)
obx = products/cod-WEEK--DOW-.obx                    ; satellite attitude         (precise)
ion = products/cod-WEEK--DOW-.ion                    ; global ionospheric map     (common)
vmf = products/VMFG_-YEAR--DOY-                      ; VMF1 grid                  (common)
```

Figure 3.6: The "product" section of configuration.

3.1.1.7 table

This section specifies the paths to the table files. An example is provided in Figure 3.7. More details can be found in Section 3.1.3. As with products, relative paths are resolved relative to the directory where pppx is executed.

```
[table]
igsatx     = ../../table/igs14.atx
oceanload = ../../table/oceanload
channel    = ../../table/glonass_chn
gpt2w      = ../../table/gpt2_1wA.grd
orography = ../../table/orography_ell
```

Figure 3.7: The "table" section of configuration.

3.1.1.8 output

This section specifies the output path and log level. An example is provided in Figure 3.8. More details can be found in Section 3.2.

```
[output]
path  = ./                        ; the folder where results will be stored
level = info                      ; opt: off/critical/error/warn/info/debug/trace
```

Figure 3.8: The "output" section of configuration.

3.1.2 Product

There are two options for satellite products in each solution mode, "brdc" and "precise", controlled by the configuration field "[product] src". The "brdc" option uses satellite orbits and clocks computed from broadcast ephemeris, whereas the "precise" option uses orbits and clocks from IGS precise products. Note that "[table] igsatx" is required when "[product] src" is set to "precise".

Here are some clarifications:

- All products labeled "precise" will not be used if "[product] src" is set to "brdc"
- Precise orbits and clocks can be used for SPP, but the Differential Code Bias (DCB) issue is not handled for single-frequency SPP; in this case, it is recommended to set "[model] iono = IF"
- Only the SP3 file is required for the "precise" option; other precise products are optional. If the CLK product is missing, the clock corrections from the SP3 file are used
- "ion" and "vmf" are only mandatory when the corresponding models are specified
- PPP specific:
  - Broadcast ephemeris can also be used for PPP; precise products are not used in this case
  - The nominal attitude is used if the OBX product is missing
  - The BIA product is used if "[estimation] ppp_ar" is set to "yes"
  - Pole tide corrections are not applied if the ERP product is missing, but this is fine for most applications

3.1.3 Table

Several table files are required when specific models are used for data processing. These mainly include the antenna model, the tropospheric model, and the tide model. A complete list of the required table files is provided in Table 3.2. The instructions for downloading each table file are provided in Appendix A.3.

Table 3.2: List of table files.

Table	Description
channel	Frequency channel numbers for GLONASS satellites
gpt2w	Coefficients of the GPT2w tropospheric model
igsatx	Satellite and receiver antenna models in the IGS ANTEX format
oceanload	Ocean tide loading coefficients for stations
orography	Ellipsoidal station heights of the VMF1 grid points

Some clarifications are provided below:

- "channel" is only needed when GLONASS carrier phase measurements are used for data processing (i.e., ppp/rtk/tdp)
- "gpt2w" must be set when "[model] trop" is set to "GPT2w"
- "igsatx" is mandatory if "[product] src" is set to "precise". When "igsatx" is specified, the satellite and station PCO corrections are applied for all solution

modes, but the satellite and station PCV corrections are applied only for PPP and RTK.

- "oceanload" is optional; ocean tide loading is not corrected if the coefficients for the processed station are missing
- "orography" is needed when "[model] trop" is set to "VMF1"

Although table files are not always required, it is recommended to properly set all table paths to simplify the process and avoid potential issues.

3.2 Output

This section introduces the output files generated by the pppx software. An overview of these files is provided in Table 3.3. The function and format of each file are described in detail in the following sections.

Table 3.3: List of output files.

Output	Description
pos file	Receiver position, clock (GPS) and ZTD estimates for each epoch
log file	Information for debugging, including cycle slip detection, a priori and post-fit residuals, and ambiguity resolution
stat file	Post-fit residuals and various estimates in the RTKLIB stat format, intended for visualization with RTKLIB

3.2.1 pos file

The primary output of pppx is the pos file, which contains the main estimates for each processed epoch. These include station coordinates in the Earth-centered and Earth-fixed (ECEF) coordinate system, standard deviations of the coordinate estimates, receiver clock errors for the GPS system, and Zenith Total Delays (ZTD).

Figure 3.9 shows the beginning of a sample pos file. The units used for coordinates, clock errors, and ZTD estimates are meters. Note that the coordinate system aligns with the satellite orbit products used for data processing. By default, the standard deviations of coordinates are not included in the output when "fgo" is selected as the solver, because of the long computation time required.

3.2.2 log file

A log file will be generated if the "[output] level" in the configuration file is not set to "off". This file contains debugging information such as cycle slip detection, outlier detection, and a-prior and post-fit residuals. Each log message generally begins with a

mjd	sod	nsat	x	y	z	stdx	stdy	stdz	rck(m)	zhd	zwd	dzwd
59679	0.00	17	4331298.023	567538.192	4633133.548	0.795	0.278	0.797	-9.8	2.081	0.025	0.2127
59679	30.00	17	4331299.398	567538.409	4633134.059	0.494	0.196	0.471	-8.9	2.081	0.025	0.1388
59679	60.00	17	4331299.590	567538.000	4633134.399	0.370	0.159	0.339	-8.4	2.081	0.025	0.0210
59679	90.00	17	4331299.938	567538.075	4633134.310	0.298	0.137	0.265	-8.2	2.081	0.025	0.0189
59679	120.00	17	4331299.981	567538.018	4633134.309	0.253	0.122	0.221	-8.1	2.081	0.025	0.0125
59679	150.00	17	4331299.899	567537.991	4633134.164	0.219	0.111	0.187	-8.0	2.081	0.025	0.0160
59679	180.00	17	4331299.918	567538.003	4633134.082	0.191	0.103	0.160	-8.0	2.081	0.025	0.0129
59679	210.00	17	4331299.932	567538.003	4633134.040	0.169	0.095	0.139	-8.1	2.081	0.025	0.0147
59679	240.00	17	4331299.774	567537.826	4633134.008	0.151	0.090	0.121	-8.1	2.081	0.025	0.0187
59679	270.00	17	4331299.776	567537.830	4633133.938	0.136	0.085	0.107	-8.2	2.081	0.025	0.0248

Figure 3.9: The screenshot of a sample pos file.

letter that indicates the message type: 'T' for Trace, 'D' for Debug, 'I' for Information, 'W' for Warning, 'E' for Error, and 'C' for Critical.

Figure 3.10 shows the screenshot of a sample log file. For example, a cycle slip was detected at epoch 86370.0 (seconds of the day). Following the PRN number, the azimuth, elevation, and between-epoch difference of the geometry-free (GF) combination of the satellite are provided. The subsequent line indicates that a new ambiguity parameter (numbered 410) was introduced at that epoch. The following lines provide information for all satellites available at that epoch. For each satellite, listed after its PRN and epoch time, these include the a priori residuals of pseudorange and carrier phase, the corresponding post-fit residuals, the ambiguity estimate in cycles, and the azimuth and elevation.

```
I ## 86370.00
I  GfJump: 86370.0 G30 -170.3 10.2  val: 16.07
I +++ G30 410 86370.00
I E04 86370.00   -0.432   -0.005     -0.423      0.004    187.118 -100.0   63.8
I E09 86370.00    3.216    0.011      3.222      0.016      0.179 -109.7   11.2
I E11 86370.00    0.231   -0.006      0.240      0.003     -4.188  -83.7   54.4
I E12 86370.00   -0.064   -0.009     -0.059     -0.004      1.739 -168.9   39.2
I E18 86370.00   -0.479   -0.016     -0.471     -0.008      3.720  -63.6   10.7
I E19 86370.00   -0.077   -0.011     -0.069     -0.003      7.753   59.7   52.5
I E21 86370.00    1.853    0.001      1.856      0.004     -2.511   93.7   14.4
I E27 86370.00    0.895   -0.015      0.901     -0.008     14.281   43.2   12.0
I E36 86370.00    0.604   -0.014      0.613     -0.005      5.998  -37.7   18.2
I G02 86370.00    1.937   -0.007      1.946      0.002     -2.125  -48.7   29.3
I G03 86370.00   -1.312    0.014     -1.309      0.017     47.787  106.7   26.2
I G04 86370.00   -0.193   -0.016     -0.185     -0.008    369.198   57.6   51.2
I G06 86370.00   -0.409   -0.012     -0.401     -0.004     54.960 -104.9   62.2
I G07 86370.00    0.904   -0.011      0.907     -0.008    -28.983  173.3   31.9
I G09 86370.00    0.800   -0.002      0.808      0.007    246.278  -20.0   86.5
I G11 86370.00   -1.175   -0.009     -1.166      0.000    -10.412  -49.6   30.7
I G19 86370.00    3.370   -0.005      3.373     -0.001     39.702 -129.2    9.7
I G26 86370.00   -1.855   -0.011     -1.850     -0.005     46.984   44.9    7.4
I G30 86370.00   -0.267   24.679     -0.266      0.000    354.394 -170.3   10.2
```

Figure 3.10: The screenshot of a sample log file.

3.2.3 stat file

A stat file is always generated for visualization with the RTKLIB software. For detailed information on the format, you can refer to the RTKLIB source code. Currently, pppx only outputs messages with the keywords "$POS", "$CLK", and "$SAT". The only difference is that the "$CLK" values are expressed in meters (instead of nanoseconds) and represent absolute clock errors for each GNSS system (rather than inter-system biases). Figure 3.11 shows a screenshot of a sample stat file.

```
$POS,2205,0.000,6,4331298.0228,567538.1921,4633133.5484,0.7955,0.2775,0.7969
$CLK,2205,0.000,6,-9.811,0.000,1.618,0.000,0.000
$SAT,2205,0.000,E03,1,-108.9,30.2,0.8635,0.0000,0,45.300,0,0,0,0,0,0
$SAT,2205,0.000,E05,1,-62.4,83.0,-0.2401,0.0000,0,52.100,0,0,0,0,0,0
$SAT,2205,0.000,E09,1,62.2,39.7,0.2750,0.0000,0,47.500,0,0,0,0,0,0
$SAT,2205,0.000,E15,1,-49.4,34.3,-0.5920,0.0000,0,47.300,0,0,0,0,0,0
$SAT,2205,0.000,E24,1,119.5,7.2,0.8040,0.0000,0,38.900,0,0,0,0,0,0
$SAT,2205,0.000,E31,1,74.2,15.2,-0.4149,0.0000,0,41.800,0,0,0,0,0,0
$SAT,2205,0.000,E36,1,177.0,24.0,-0.3656,0.0000,0,43.300,0,0,0,0,0,0
$SAT,2205,0.000,G02,1,-47.8,28.1,1.0852,0.0000,0,44.700,0,0,0,0,0,0
```

Figure 3.11: The screenshot of a sample stat file.

3.3 Visualization

This section explains how to visualize the solutions generated by pppx.

3.3.1 With Python

A Python script named "plot_ppppos.py" is included in the software package and is located in the "scripts/" folder. This script can generate plots of either (1) the positioning time series in the east, north, and up components (relative to the mean position) or (2) additional time series of the receiver clock and ZTD estimates. The general usage of the script is demonstrated with the commands below, and the corresponding figures are shown in Figure 3.12:

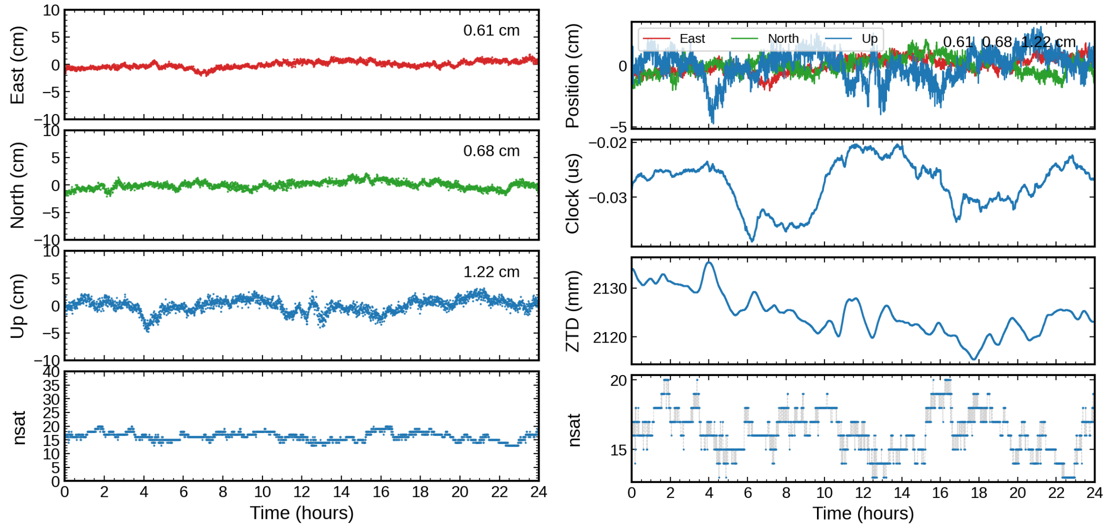
```
# (1) plot position estimates only
$ ../../scripts/plot_ppppos.py pos_file -s

# (2) plot position, receiver clock, and ZTD estimates
$ ../../scripts/plot_ppppos.py pos_file -a

# Options:
# -a  Plot position, receiver clock, and ZTD estimates
# -i  Interactive mode
# -s  Fixed scale for the y-axis; otherwise, the scale is set automatically
```

3.3.2 With RTKLIB

The RTKLIB software is recommended for a richer interactive view of the various plots and for visualizing post-fit residuals. One limitation is that the author of RTKLIB provides executables only for Windows; Linux users must compile the GUI applications themselves using the Qt library.

Using RTKLIB for visualization is straightforward:

1. Open the application "rtkplot.exe"
2. Click the menu "File" > "Open Solution-1" > Select the generated stat file

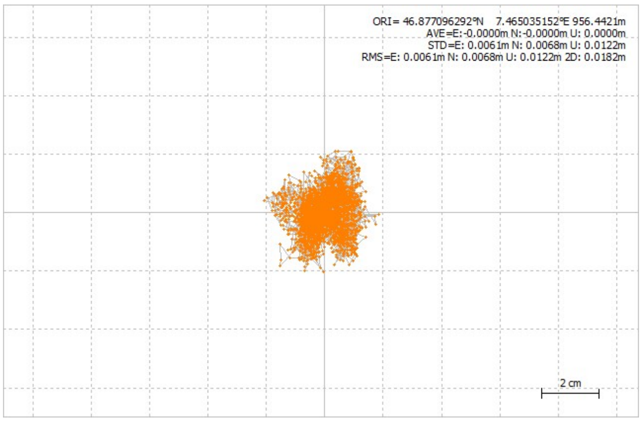


(a) Plotted with option "-s" (b) Plotted with option "-a"

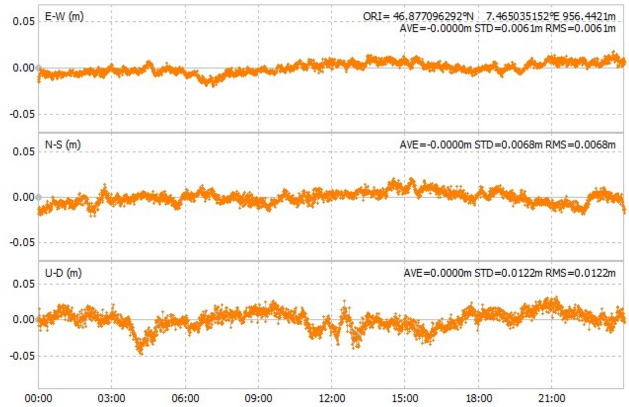
Figure 3.12: Visualization of an FGO-based PPP solution with the Python script.

3. Interactively view the various plots

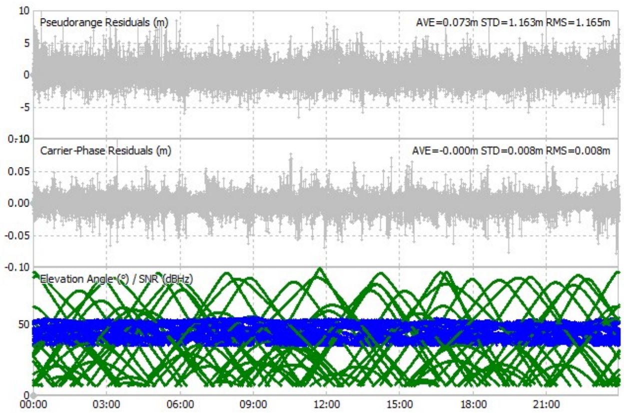
Figure 3.13 shows some example visualizations. A useful tip: you can open two solutions within the same window for easy comparison.



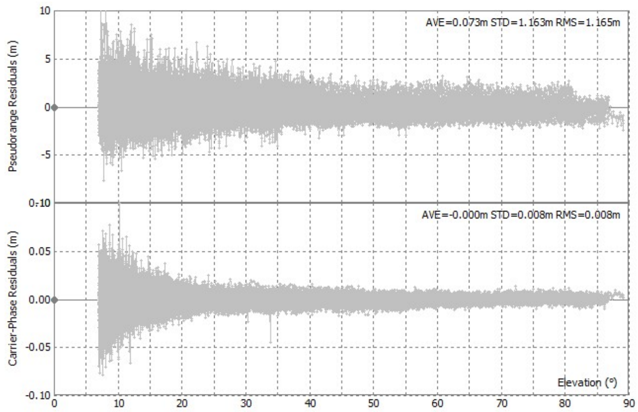
(a) Ground track plot.



(b) Positioning time-series plot.



(c) Residual time-series plot.



(d) Residual-elevation plot.

Figure 3.13: Visualization of an FGO-based PPP solution with RTKLIB.

4 Examples

This section provides examples of using pppx for various types of data processing. The corresponding folder is 'example/pppx/'. Specifically, two RINEX files and some products are provided in "example/pppx/rinex/" and "example/pppx/products/", respectively. Basic information on the observation and product files can be found in Tables 4.1 and 4.2, respectively. Note that the VMF1 grid product is a file splicing the 00 h, 06 h, 12 h, and 18 h epochs of the current day with the 00 h epoch of the next day. By default, GPS and Galileo observations are used in each example, except for the RTK example, which is GPS-only.

Table 4.1: GNSS data used in examples.

Data	Description
ZIM200CHE_R_20221000000_01D_30S_MO.rnx	Rover station (GREC)
ZIMM00CHE_R_20221000000_01D_30S_MO.rnx	Reference station (G)
ALGO00CAN_R_20221000000_15M_01S_MO.rnx	High-rate data

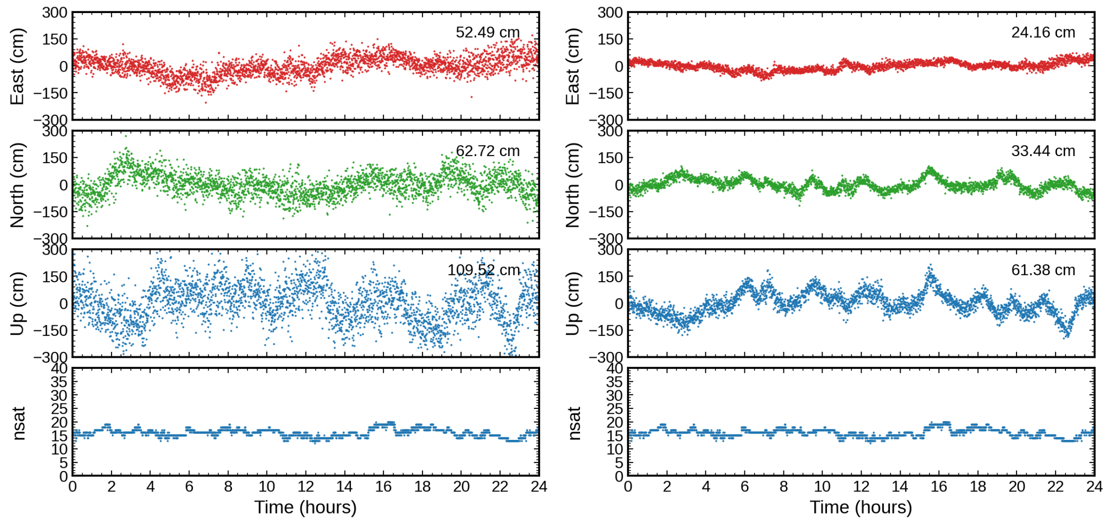
Table 4.2: Products used in examples.

Product	Description
BRDC00IGS_R_20220010000_01D_MN.rnx	Broadcast ephemeris for multi-GNSS
COD0MGXFIN_20221000000_01D_05M_ORB.SP3	Precise orbit from CODE
COD0MGXFIN_20221000000_01D_30S_CLK.CLK	Precise clock from CODE
COD0MGXFIN_20221000000_01D_15M_ATT.OBX	Satellite attitude from CODE
COD0MGXFIN_20221000000_01D_01D_OSB.BIA	Satellite bias from CODE
COD0MGXFIN_20221000000_03D_12H_ERP.ERP	ERP from CODE
codg1000.22i	GIM from CODE
VMFG_2022100	spliced VMF1 grid from TU Vienna

4.1 SPP: Single Point Positioning

The pppx software supports multi-GNSS SPP, including GPS, GLONASS, Galileo, BeiDou, and QZSS, with broadcast ephemeris or precise satellite products. Various tropospheric and ionospheric models are supported. However, only the kinematic mode is currently supported, with either LSQ or FGO as the solver. Note that LSQ- and FGO-based SPP are both epoch-wise solutions (i.e., with no constraints between epochs); the only difference is that FGO uses the Huber loss function (2 m threshold) to improve robustness. Here, the LSQ solver is used, as the GNSS data in the examples are from geodetic-grade receivers.

Here, SPP solutions using the IF combination and single-frequency observations are demonstrated. Both examples use broadcast ephemeris, which is the most common



(a) IF solution

(b) SF solution

Figure 4.1: Positioning results of SPP solutions.

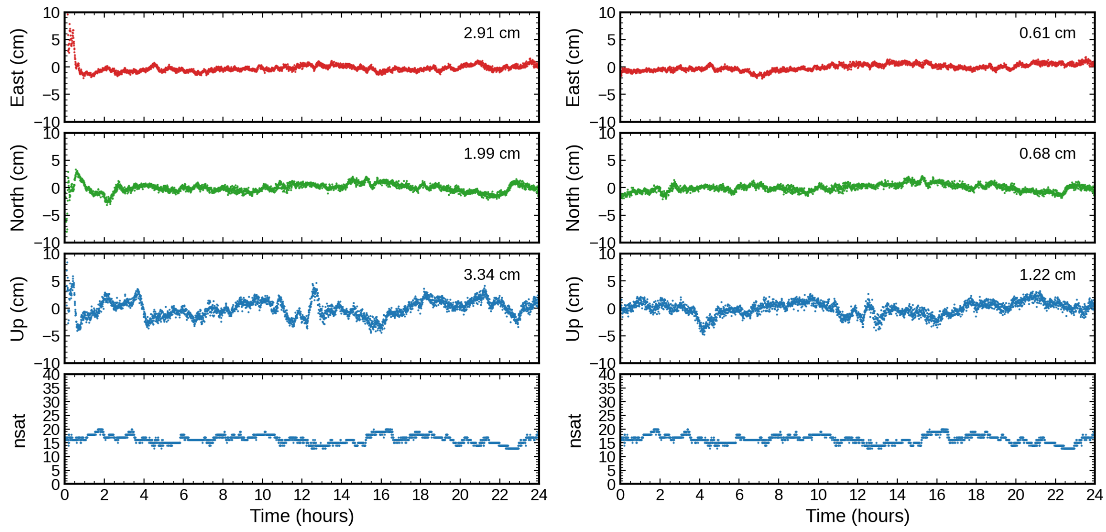
case for SPP users. Specifically, the GMF model and the IF combination are used to correct tropospheric and ionospheric delays for the IF solution, while the GMF model and a GIM product are used for the single-frequency solution. The other settings are kept at their defaults. The corresponding configuration files can be found as "example/pppx/00_spp_if.ini" and "example/pppx/01_spp_sf.ini", respectively.

The positioning results are shown in Figure 4.1. The single-frequency solution is better than the IF solution. This is because the IF combination amplifies pseudorange noise by about a factor of three, while the ionospheric delays can be well corrected with the GIM product for station ZIM2, which is located at mid-latitude.

4.2 PPP: Precise Point Positioning

Here, PPP solutions are demonstrated with the Kalman filter or FGO as the solver. The corresponding configuration files can be found as "example/pppx/02_ppp_ekf.ini" and "example/pppx/03_ppp_fgo.ini", respectively. The Kalman-filter-based solution simulates the real-time scenario and outputs a forward-only solution. The FGO-based solution is dedicated to post-processing and outputs a batch solution, i.e., one without a convergence period. In addition, FGO uses the Huber loss function to improve robustness, with thresholds of 2 m and 0.02 m for pseudorange and carrier phase observations, respectively.

The positioning results are shown in Figure 4.2. There is no convergence period for the FGO-based solution, whereas the EKF solution takes approximately 15 min to converge. This is because FGO uses all the GNSS observations to form a single large LSQ problem and solves all the parameters jointly, whereas EKF solves the parameters epoch by epoch.



(a) EKF solution (b) FGO solution

Figure 4.2: Positioning results of PPP solutions.

4.3 RTK: Real-Time Kinematic

The pppx software only supports short baseline solutions. Thus, "[model] trop" and "[model] iono" are always set to "none". Both EKF and FGO are supported as solvers for RTK. As before, the EKF solution simulates the real-time scenario and outputs a forward-only solution, while the FGO solution is dedicated to post-processing and outputs a batch solution without a convergence period. The same Huber loss function as in PPP is adopted for FGO-based RTK processing. Here, we show the RTK solution computed with EKF as an example. The corresponding configuration file can be found as "example/pppx/06_rtk.ini".

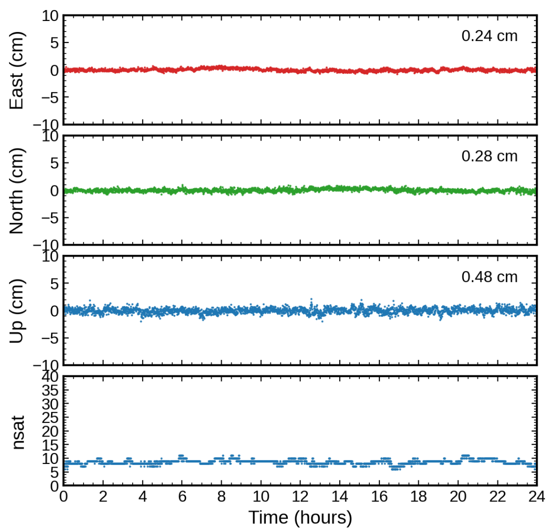


Figure 4.3: Positioning results of a short baseline solution between ZIM2 and ZIMM.

The positioning result is shown in Figure 4.3. The precision is 0.24 cm, 0.28 cm, and 0.48 cm for the east, north, and up components, respectively. These are typical precision

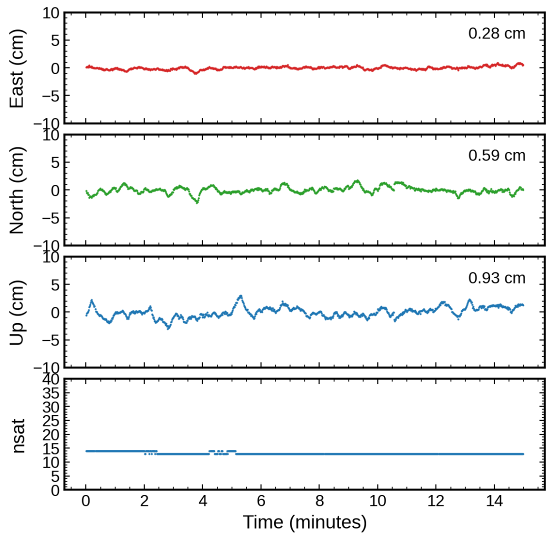


Figure 4.4: TDP positioning results for the ALGO station.

values for short baseline solutions with resolved ambiguities.

4.4 TDP: Time-Differenced Positioning

TDP is useful for earthquake studies because it has no convergence time and achieves high precision over a short time span. Ambiguities are eliminated by time-differencing, and the station displacements between adjacent epochs are estimated. The displacements are then accumulated onto the initial station position, and the absolute position for each epoch is written to the pos file.

An example is provided as "07_tdp", where the 1-Hz GNSS data collected at the ALGO station are processed with TDP. Specifically, precise satellite products are used instead of broadcast ephemeris. In this way, the drift in the positioning time series caused by error accumulation becomes much smaller. As ALGO is a static station, the ground truth displacement is zero. The TDP positioning precisions over 15 min are 0.28 cm, 0.59 cm, and 0.93 cm in the east, north, and up components, respectively.

5 Troubleshooting

Text

6 FAQs

Text

A File Download

A.1 Download GNSS observations

1. CDDIS: https://cddis.nasa.gov/archive/gnss/data
2. EPN: ftp://ftp.epncb.oma.be/pub/obs
3. PBO: https://data.unavco.org/archive/gnss/
4. GA: https://data.gnss.ga.gov.au/docs/home/gnss-data.html
5. Curtin: http://saegnss2.curtin.edu/ldc
6. Hongkong: https://rinex.geodetic.gov.hk

A.2 Download IGS Products

The following HTTP sites are available for downloading GNSS products:

1. CDDIS: https://cddis.nasa.gov/archive/gnss/
2. CODE: http://ftp.aiub.unibe.ch/CODE_MGEX/CODE/
3. ESA: http://navigation-office.esa.int/products/gnss-products/
4. CNES: http://www.ppp-wizard.net/daily.html

FTP sites:

1. CDDIS: ftps://gdc.cddis.eosdis.nasa.gov/gnss/
2. CODE: ftp://ftp.aiub.unibe.ch/CODE_MGEX/CODE/
3. WHU: ftp://igs.gnsswhu.cn/pub/
4. IGN: ftp://igs.ign.fr/pub/igs

Other products:

1. VMF1: http://vmf.geo.tuwien.ac.at/trop_products/GRID/2.5x2/VMF1/VMF1_OP/

A.3 Download Table Files

A.3.0.1 Ocean tide loading

Old portal: http://holt.oso.chalmers.se/loading/index-aside-2404271219.html

New portal: `https://barre.oso.chalmers.se/loading/l.php`

Here, we use the old portal to download oceanload coefficients.

1. Ocean tide model -> select **"FES2004"**
2. Keep other options as default values
3. Enter the station list: either XYZ or BLH
4. Enter an email address and click **"Submit"**

The required oceanload coefficients will be sent to you by email; you then need to append them to the table file "table/oceanload".

A.3.0.2 ANTEX

`https://files.igs.org/pub/station/general`